

→ Calcul Numérique.

$$f: \mathbb{R} \rightarrow \mathbb{R}$$

$$x \mapsto f(x) = y$$

Soit f une fonction

fame

$$f(x) \quad f \leftarrow x \in \mathbb{R}$$

Soit f une fonction

$$\text{model. predict } f_{\theta}(x) = \hat{y}$$

$$NN \rightarrow L(\theta) \leftarrow 0 \quad \begin{cases} f(x+h) = 2x+h \\ f(x) = 2x \\ 2x+h-2x = \alpha h \end{cases}$$

$$f: I \rightarrow \mathbb{R} \quad x \mapsto f(x); \quad f'(x_0) = \lim_{h \rightarrow 0} \frac{f(x_0+h) - f(x_0)}{h}$$

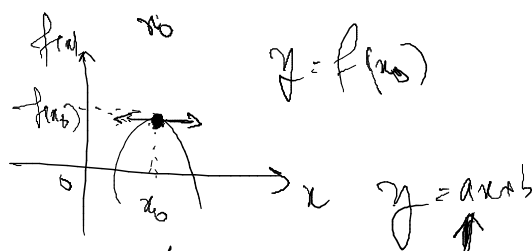
$x_0 \in \mathbb{R} \quad f'(x_0) \rightarrow \text{nombre dérivé}$

$$f(x) \quad f(x) = 2x \quad f'(x) = 2x$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \rightarrow \infty$$

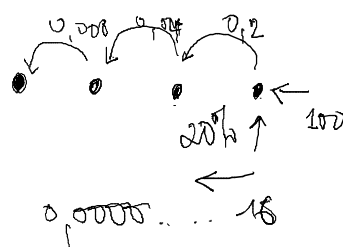
$$f'(x) \rightarrow f''(x) \rightarrow \text{note dérivée}$$



$$(\lambda f + g)'(x) = \lambda f'(x) + g'(x)$$

$$[\lambda f(x) + g(x)]'$$

$$\frac{d}{dx} [\lambda f(x) + g(x)]$$



$$f: \mathbb{R}^n \rightarrow \mathbb{R}, \quad n \geq 2$$

$$x_1, x_2, \dots, x_n \rightarrow f(x_1, \dots, x_n)$$

$$\frac{\partial f}{\partial x_i}(x) = \lim_{h \rightarrow 0} \frac{f(x_1, \dots, x_i+h, \dots, x_n) - f(x)}{h}$$

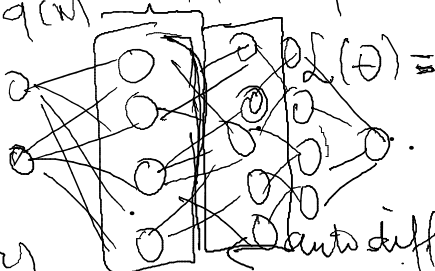
$$f(x,y) \quad \left| \quad \frac{\partial f}{\partial x}$$

$$f(x,y) = x^2 y + \sin(xy) \quad \left| \quad \frac{\partial f}{\partial x}(x,y) = 2xy + \cos(xy)$$

$$\sigma(x) = \frac{1}{1 + e^{-x}} \quad \left| \quad g \circ f(x) = g(f(x))$$

$$g(x) = \frac{1}{x} \quad \text{et} \quad f(x) = 1 + e^{-x}$$

$$1 - \sigma(x) = \frac{e^{-x}}{1 + e^{-x}}$$



$$\sigma(x)$$

$$\frac{\partial L}{\partial \theta} \quad L \rightarrow f(\theta)$$

$$\sigma'(x) = \sigma(x) [1 - \sigma(x)]$$

$$\frac{\partial f}{\partial x}(x,y) = 2xy + \cos(xy)$$

↑ $x, y \rightarrow \text{symbols}$

autograd // differentiation automatique.

$$f(x) = \left(\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n} \right) (x_0, y_0, \dots)$$

$$\begin{aligned} f: \mathbb{R}^n &\rightarrow \mathbb{R}^m \\ g: \mathbb{R}^m &\rightarrow \mathbb{R}^p \end{aligned} \quad \left| \quad \begin{aligned} h &= g \circ f \\ x &\in \mathbb{R}^n \end{aligned} \right.$$

$$f(x+h) = f(x) + J_f(x)h + o(\|h\|)$$

$$y = f(x), \quad k = J_f(x)h + o(h)$$

$$h(x+h) = \underbrace{g(f(x))}_{y} + J_h(x) \cdot h + o(h)$$

$$\underline{h(x+h) = g(y) + \underbrace{J_g(y)}_{J_g(f(x))} \cdot \underbrace{J_f(x)}_{J_f(x)} h}$$

Graphes de Calcul

$$z = f(y) \quad y = g(u)$$

$$f(y) = y^2; \quad g(u) = \sin u$$

$$\frac{dz}{du} = \frac{dz}{dy} \cdot \frac{dy}{du} = \frac{dz}{du}$$

$$\frac{dz}{du} = 2y \cdot \cos u \quad \left| \quad \begin{aligned} z &= f(y) = y^2 \\ &= [g(u)]^2 \end{aligned} \right.$$

$$\begin{aligned} \frac{dz}{du} &= 2 \sin u \cdot \cos u \\ &= \sin(2u) \end{aligned} \quad \left| \quad \begin{aligned} z(u) &= \sin^2 u \end{aligned} \right.$$

Gradient: $f: U \rightarrow \mathbb{R}, \quad x \in \mathbb{R}^n$

$$\nabla f(x) = \begin{pmatrix} \frac{\partial f}{\partial x_1}(x) \\ \vdots \\ \frac{\partial f}{\partial x_n}(x) \end{pmatrix} \in \mathbb{R}^n$$

$$\begin{aligned} f &= (f_1, \dots, f_m): \mathbb{R}^n \rightarrow \mathbb{R}^m, \quad x \in \mathbb{R}^n \\ &\quad \downarrow \\ &\quad \mathbb{R}^n \rightarrow \mathbb{R}^m \\ &\quad m \times n \end{aligned}$$

$$J_f(x) = \begin{pmatrix} \frac{\partial f_1}{\partial x_1}(x) & \dots & \frac{\partial f_1}{\partial x_n}(x) \\ \vdots & & \vdots \\ \frac{\partial f_m}{\partial x_1}(x) & \dots & \frac{\partial f_m}{\partial x_n}(x) \end{pmatrix}$$

$$f: \mathbb{R} \rightarrow \mathbb{R}^m$$

$$J_f(x) = \left(\frac{\partial f}{\partial x_1} \dots \frac{\partial f}{\partial x_n} \right)^T$$

$$J_f(x) = \begin{pmatrix} \frac{\partial f}{\partial x_1} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{pmatrix}$$

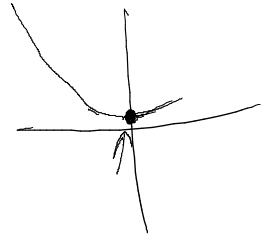
$$\nabla^2 f(x) = \begin{pmatrix} \frac{\partial^2 f}{\partial x_1^2} & \dots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \vdots & & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \dots & \frac{\partial^2 f}{\partial x_n^2} \end{pmatrix} \quad f \in \mathcal{C}^2$$

$$\frac{\partial^2 f}{\partial x_1^2}, \frac{\partial^2 f}{\partial x_1 \partial x_2}, \dots, \frac{\partial^2 f}{\partial x_n \partial x_1}, \dots, \frac{\partial^2 f}{\partial x_n^2}$$

$$f \in \mathcal{C}^2 \mid \text{continuité des dérivées}$$

$$\nabla^2 f(x) = \begin{pmatrix} \frac{\partial^2 f}{\partial x_1^2} & \dots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \vdots & & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \dots & \frac{\partial^2 f}{\partial x_n^2} \end{pmatrix}$$

$$\frac{\partial f}{\partial x} \rightarrow \text{fonction}$$



$$f(x) = x^2 \mid f'(x) = 2x \mid f''(x) = 2$$

$$f(x, y) = x^2 - y^2$$

$$f(x, y) \rightarrow$$

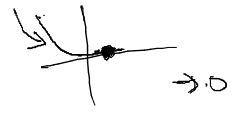
$$(x, y) = (0, 0)$$

$$\begin{cases} \frac{\partial f}{\partial x} = 0 \\ \frac{\partial f}{\partial y} = 0 \end{cases}$$

$$\begin{cases} 2x = 0 \\ -2y = 0 \end{cases} \Rightarrow (0, 0)$$

$$\begin{cases} \text{Convexe} \\ \text{Concave} \\ \text{ni convexe, ni concave} \end{cases}$$

$$y = ax^2 + bx + c$$



$$\theta_{k+1} = \arg \min_{\theta} J(\theta)$$

$$J(\theta) = \theta^2 \Rightarrow \theta^* = 0$$

$$\nabla J(\theta) = 2\theta \mid \theta = 0 \Rightarrow \nabla J(\theta) = 0$$

$$(2.5) \quad f(x+h) = f(x) + f'(x)h + \frac{1}{2} f''(x)h^2 + \dots$$

$$J(\theta + \Delta\theta) \approx J(\theta) + \nabla J(\theta)^T \Delta\theta$$

$$\nabla J(\theta)^T \Delta\theta \mid \nabla J(\theta) = \begin{pmatrix} \frac{\partial J}{\partial \theta_1} \\ \vdots \\ \frac{\partial J}{\partial \theta_n} \end{pmatrix}$$

$$a^T \cdot b = \|a\| \cdot \|b\| \cos(\alpha) \mid \alpha = \widehat{a, b}$$

$$\nabla J(\theta)^T \cdot \Delta\theta = \|\nabla J\| \cdot \|\Delta\theta\| \cos \alpha \mid \frac{\partial}{\partial \eta}$$

$$\Delta\theta = -\eta \nabla J(\theta)$$

$$\Delta\theta = -\eta(N) \nabla J(\theta)$$

$$J(\theta + \Delta\theta) \approx J(\theta) + \nabla J(\theta)^T (-\eta \nabla J(\theta))$$

$$\text{Note: } A^T A = \|A\|^2$$

$$J(\theta + \Delta\theta) \approx J(\theta) - \eta \nabla J(\theta)^T \nabla J(\theta)$$

$$J(\theta + \Delta\theta) \approx J(\theta) - \eta \|\nabla J(\theta)\|^2$$

$$J(\theta + \Delta\theta) - J(\theta) \approx -\eta \left\| \nabla J(\theta) \right\|^2$$

$$f'(x_0) = \lim_{h \rightarrow 0} \frac{f(x_0+h) - f(x_0)}{h}$$

$$J(\theta + \Delta\theta) \approx J(\theta) - \eta \|\nabla J(\theta)\|^2$$

$$\Delta\theta = -\eta \nabla J(\theta) \quad \parallel \quad \Delta\theta$$

$$\theta_{\text{new}} = \theta_{\text{ancien}} + \Delta\theta$$

$$\theta_0 \quad \theta_1$$

$$\theta_{\text{new}} = \theta_{\text{ancien}} - \eta \nabla J(\theta)$$

For $k \in \mathbb{N}$,

$$\theta_{k+1} = \theta_k - \eta \nabla J(\theta_k)$$

↑
iteration k

$$J(\theta + \Delta\theta) < J(\theta)$$

$$-\eta \|\nabla J(\theta)\|^2 < 0$$

$$J(\theta + \Delta\theta) - J(\theta) < 0$$

$$J(\theta + \Delta\theta) < J(\theta)$$

$$f(x) = \frac{1}{2} x^T A x \quad \parallel \quad \frac{1}{2} A \|x\|^2$$

$$\nabla f(x) = A x, \quad \nabla^2 f(x) = A$$

$$L = \max_i \lambda_i; \quad \alpha = \frac{1}{L}$$

$$\alpha = 0,01$$

